of the magnetic field depends on the direction of the current flow.

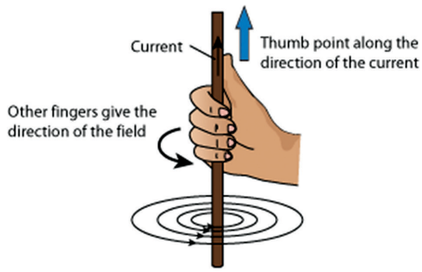


Fig 3.7 *Right Hand Grip Rule*

Right Hand Grip Rule

- It is used to determine the direction of the magnetic field in a current carrying conductor.
- If you wrap your fingers around the wire with your thumb pointing direction of current flow, your index finger will point the direction of magnetic field.

Right Handed Cork Screw Rule

- Assume a right handed cork screw to be along the wire to advance in the direction of current.
- The motion of handle gives the direction of magnetic lines around the conductor.

Force Between Parallel Conductors

When two current carrying conductors are parallel to each other, a mechanical force act on each conductor. This force is due to magnetic field produced in the two conductors. If the currents are in the same direction, the forces are attractive. If the currents are in the opposite direction, the forces are repulsive.

i) Current in The Same Direction

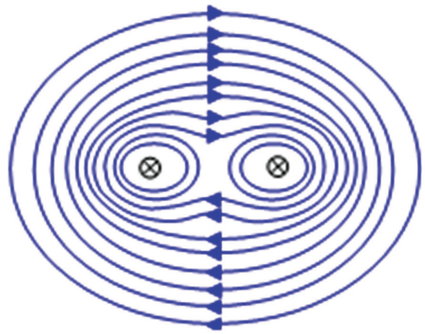


Fig 3.8 *Currents in Same Direction*

- If two wires (A, B) carrying current in same direction are brought together, their magnetic fields will aid one another and attracts.
- Since the flux lines around two conductors are going in the same direction, the flux lines join and the field brings the wire together.

ii) Current in The Opposite Direction

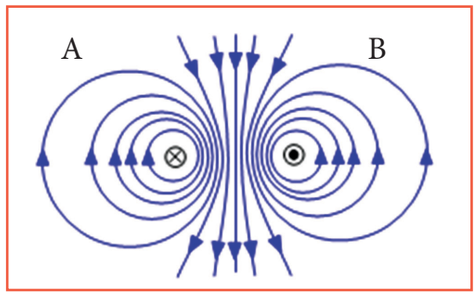


Fig 3.9 *Currents in Opposite Direction*

- If two wires (A, B) carrying current in opposite directions are brought together their magnetic field will oppose one another.
- Since the flux lines around two conductors are going in the opposite direction, the flux lines cannot cross and the field moves the wires apart.

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